Scan2Work — pipeline

From a splat scan to a named, routable twin

Two routes: local-first and portable, or online with Gemini for stronger naming.



Two routes through the same pipeline · one switch: VALMET_SEGMENTER

Route 1 · local

Local-first

Portable. Open models on one machine: no API key, and no plant image ever leaves the site.

SAM 2.1 · SigLIP 2 · DINOv2 · OWLv2
485 objects · Apache-2.0 weights only

Route 2 · online

Gemini

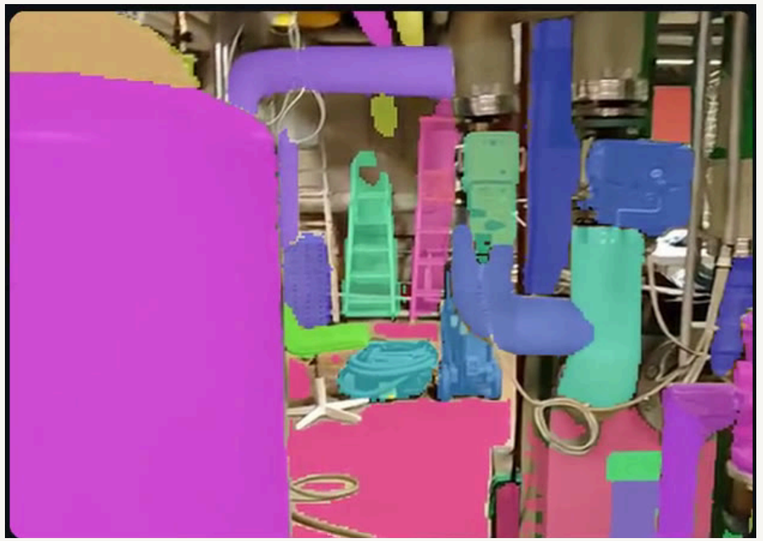
More powerful at naming. Geometry finds the objects; Gemini names each one from its real photo.

Gemini 3.8 Flash
85% naming accuracy · 467 objects

Route 1 · local-first, portable

Local-first: segment everything, then recognise

01 Align	COLMAP, 334 video frames on the splat
02 Segment	SAM 2.1 hiera-large, every frame
03 Stitch	masks → Gaussians, view consensus
04 Name	SigLIP 2 so400m + DINOv2 base
05 Check	OWLv2 boxes as a second opinion
06 Classify	linear probe, active learning
07 → 08	pipes, bind, route, diagnose: shared



SAM 2.1 cuts every frame into pieces, no prompts and no word list. Pieces join in 3D only where most views agree.

VALMET_SEGMENTER=local · Apache-2.0 weights only, SAM 3 and YOLO excluded by licence · plant images stay on site

Route 2 · online

Online: geometry segments, Gemini names

01 Ingest	PLY decode, level floor to y = 0
02 Segment	colour + geometry rules, DBSCAN
03 Name	Gemini 3.8 Flash, real photo + 3D box
04 Merge	Gemini detection, 42 frames → 3D
05 Pipes	normal voting, cylinder fit
06 Bind	30 workbook devices, class + position
07 Route	A*, 5 cm grid, 0.15 m robot
08 Diagnose	EmbeddingGemma 300M + gpt-6-luna

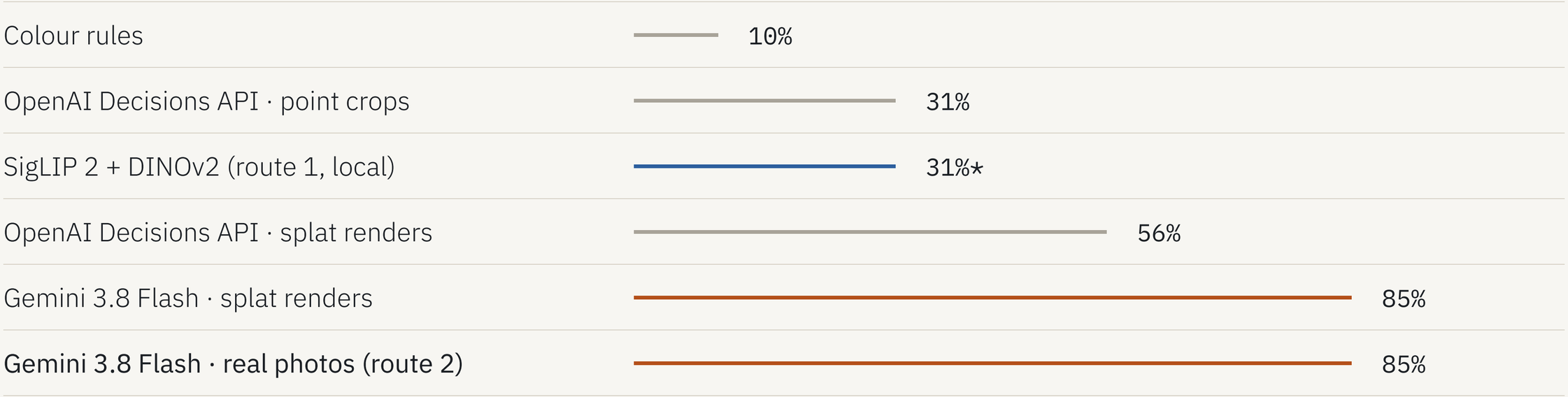


What Gemini sees: the sharpest real frame, the object’s 3D box drawn in. Answer: “control valve” (CV-003).

orange = hosted model · fallback: OpenAI Decisions API · all answers cached, re-runs need no key

Naming · same 39 hand-labelled objects

Same images, different model: 56% → 85%



tests/gold_labels.json · tests.scorecard, 7 Nov 2026 · * sample drawn from route 2's clusters

Navigation · pipeline/navigation.py

A collision-free route from the scan itself

Occupancy	2.5D grid, 5 cm cells, splat density
Height band	solid between 0.16 m and 1.62 m
Robot	obstacles dilated by 0.15 m radius
Search	A*, collision-checked shortcuts
Stand-off	stops 0.52 m from the target box
Proximity	assets within 2.5 m framed en route

0
samples off the floor

0.234 m
minimum clearance

1.8%
of a real walk on obstacles

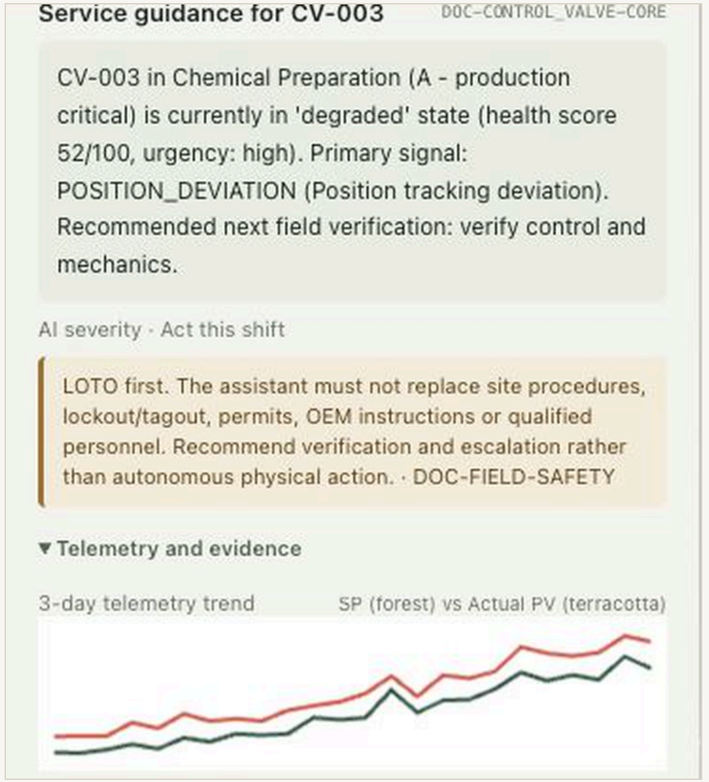


tests.scorecard · real walk = the 334 camera poses of the challenge video, checked against our obstacle map

Excel mapping · pipeline/knowledge.py

Every number on screen comes from the workbook

Source	Valmet builder pack, 30 devices
Binding	by class and position, one object each
Target	CV-003 → OBJ-206, checked on video
Carried	status, telemetry, alarms, history
Unmapped	framed and named, no borrowed data
Check	30/30 bound, 0 telemetry mismatches



procedural mapping, as the brief allows · tests.scorecard checks every shown value against the Excel pack

AI diagnosis · pipeline/diagnostics.py, retrieval.py

Grounded triage, tested blind on 13 faults

retrieve: EmbeddingGemma 300M over workbook notes → triage: gpt-6-luna ranks root causes → LOTO-first steps, each citing Excel evidence IDs

9/13
top-1, telemetry only

11/13
top-3, telemetry only

2/13
baseline: class's most common issue

tests.eval_triage · condition, health and issue codes hidden · Gemini not evaluated on triage